

Module 09: Page Fault Handling and EAT Performance

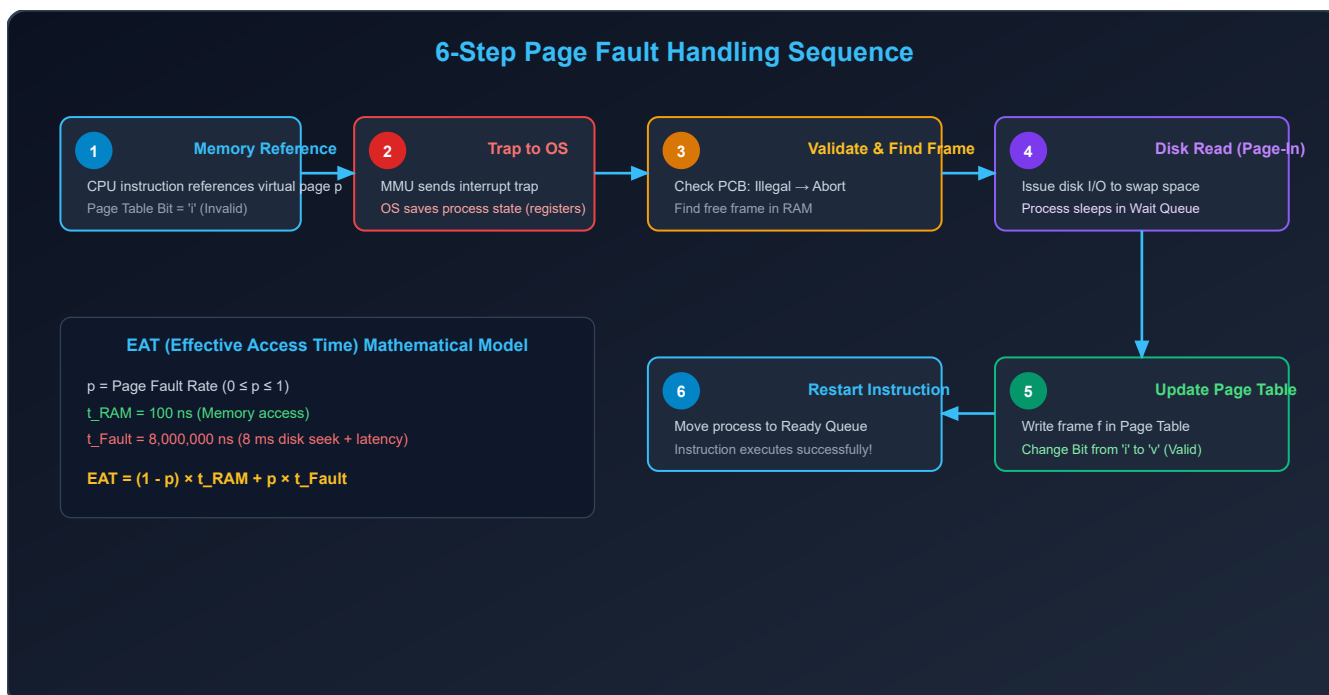
AKTU BCS401

Unit 4

Fault Recovery & EAT

1. Page Fault 6-Step Sequence

1. **Memory Reference:** CPU accesses invalid page → Hardware MMU trap.
2. **Trap to OS:** Kernel saves process state to PCB.
3. **Legality Check:** If legal, find free frame; if illegal, abort process.
4. **Disk I/O Swap-In:** Read missing page from swap space into RAM frame.
5. **Update Page Table:** Enter frame number, flip bit to 'v' (Valid).
6. **Restart Instruction:** Trapped instruction restarts seamlessly.



2. Effective Access Time (EAT) Formula

$$EAT = (1 - p) \times t_{RAM} + p \times (\text{Page Fault Service Time})$$

Solved Example: $t_{RAM} = 200 \text{ ns}$, Fault Service Time = 8 ms.

10% max slowdown ke liye: $p \leq 20 / 8,000,000 = 1 / 400,000$ (0.00025%).

Har 4 lakh memory accesses me strictly 1 se kam page fault aana chahiye!